

THE PROPER THREE-COLOURING ZERO SET OF GENERALIZED CHORDED CYCLES IS $\{7, 8, 12, 16\}$

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ABSTRACT. Let $C_n^{(3)}$ be the cycle on $\mathbb{Z}/n\mathbb{Z}$ with all offset-three chords and, for even n , all diameter edges. We prove that the number of proper three-colourings of $C_n^{(3)}$ vanishes exactly for $n \in \{7, 8, 12, 16\}$, for every $n \geq 6$. The positive direction uses two colour blocks, $A = 01$ and $B = 21202$, in four explicit infinite families. The negative direction encodes adjacent colour differences by signs. Offset-three chords forbid three consecutive equal signs; a run bound and a sliding half-window argument then exclude precisely the four exceptional orders. The proof is elementary and does not use a transfer matrix or an asymptotic positivity estimate. Exact verification code and serialized block data are included as a diagnostic supplement.

1. INTRODUCTION

The sequence A383733 records the number $a(n)$ of proper three-colourings of a generalized chorded cycle [4]. The graph is obtained from a cycle by adding the chords of offset three and, at even orders, the diameters. Lopez-Bonilla, Allagan, Langley, and Clinton derived a closed form for the odd branch and developed transfer-matrix methods for the family [3]. The published table contains zeros at 7, 8, 12, 16, but it does not establish that no later zero occurs. A later public computational artifact checked the same zero set through $n = 3000$ and posed the all-order classification separately [1, 2].

We settle that classification. The proof is constructive except at the four exceptional orders. In particular, it avoids a subtlety of the even branch: a diameter has length depending on n , so a local transfer matrix needs a global seam condition. Instead, two short colour blocks give proper colourings at every nonexceptional order. A difference-word reduction then gives a uniform obstruction for 8, 12, 16, and a still shorter run-count obstruction for 7.

The novelty search for this note covered the OEIS record, the journal and arXiv versions of the source paper, DOI/Crossref and OpenAlex records, the SciNet problem and code artifact, and exact-phrase web queries, through

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August 29, 2026. No prior all-order proof or counterexample was found in that recorded scope. This is a bounded literature-search statement, not a claim that unindexed work cannot exist.

2. DEFINITION AND A DIFFERENCE-WORD REDUCTION

Definition 2.1. For $n \geq 6$, the graph $C_n^{(3)}$ has vertex set $\mathbb{Z}/n\mathbb{Z}$ and edge set

$$\{\{i, i+1\}, \{i, i+3\} : i \in \mathbb{Z}/n\mathbb{Z}\},$$

together with $\{\{i, i+n/2\} : i \in \mathbb{Z}/n\mathbb{Z}\}$ when n is even. Repeated unordered edges are identified. Let $a(n)$ be the number of maps $c : \mathbb{Z}/n\mathbb{Z} \rightarrow \mathbb{Z}/3\mathbb{Z}$ whose endpoints have different colours on every edge.

The convention at $n = 6$ deserves mention: offset three and the diameter coincide, exactly as in the OEIS definition [4].

Lemma 2.2 (difference-word criterion). *Let $d \in \{+1, -1\}^n$, read cyclically. It is the adjacent-difference word of a proper three-colouring of $C_n^{(3)}$ if and only if*

- (1) $\sum_i d_i \equiv 0 \pmod{3}$;
- (2) *no three cyclically consecutive signs in d are equal; and*
- (3) *if $n = 2m$, every cyclic length- m window of d has sum nonzero modulo three.*

Proof. Given a proper colouring, put $d_i = c(i+1) - c(i) \in \mathbb{Z}/3\mathbb{Z}$. The cycle edges make each difference nonzero, so its unique integer representative is $+1$ or -1 . Closing the colours around the cycle is condition (1). The colour difference across the offset-three chord from i is

$$c(i+3) - c(i) = d_i + d_{i+1} + d_{i+2} \pmod{3}.$$

A sum of three signs is divisible by three exactly when all three signs are equal, which proves condition (2). When $n = 2m$, the difference across a diameter is the sum of the corresponding length- m window, giving condition (3).

Conversely, choose $c(0) \in \mathbb{Z}/3\mathbb{Z}$ and define successive colours by $c(i+1) = c(i) + d_i$. Condition (1) closes the colouring, and conditions (2)–(3) make every chord and diameter proper. Thus no information is lost. $\square$

3. EXPLICIT COLOURINGS AT EVERY NONEXCEPTIONAL ORDER

Write colour words over $\{0, 1, 2\}$, read cyclically, and put

$$A = 01, \quad B = 21202.$$

Proposition 3.1 (four constructive families). *The following cyclic words are proper three-colourings:*

$$(4) \quad \begin{array}{ll} n \geq 9 \text{ odd} & : A^{(n-5)/2}B, \\ n \equiv 2 \pmod{4} & : A^{n/2}, \\ n = 20 & : A^3BA^2B, \\ n = 4k \geq 24 & : A^{k-1}BA^{k-4}B. \end{array}$$

Consequently $C_n^{(3)}$ is three-colourable for every $n \geq 6$ outside $\{7, 8, 12, 16\}$.

Proof. First ignore diameters. An A -string alternates, so colours at offsets one and three differ. Inside B , adjacent symbols differ, and its two internal offset-three comparisons are $2 \neq 0$ and $1 \neq 2$. Every junction in the first, third, and fourth words in (4) has one of the two local contexts

$$\cdots 0101 \mid 21202 \cdots, \quad \cdots 21202 \mid 0101 \cdots.$$

Across the two junctions, the respective triples of new offset-three comparisons are

$$((1, 2), (0, 1), (1, 2)) \quad \text{and} \quad ((2, 0), (0, 1), (2, 0)).$$

The adjacent symbols at both junctions also differ. All separating A -blocks have exponent at least two, including at the cyclic seam. Hence every cycle and offset-three edge is proper.

If $n \equiv 2 \pmod{4}$, the word $A^{n/2}$ alternates. The offsets one and three are odd, as is the diameter $n/2$, so all three edge types have differently coloured endpoints. This row includes $n = 6$.

For $n = 20$, splitting the third word into its two length-ten halves gives

$$A^3BA^2B = \underbrace{0101012120}_X \underbrace{2010121202}_Y.$$

The two displayed strings differ in every coordinate, which checks all ten diameters.

Finally let $n = 4k$, $k \geq 6$. Splitting the fourth word at its diameter gives

$$A^{k-1}BA^{k-4}B = \underbrace{A^{k-1}21}_X \underbrace{202A^{k-4}B}_Y, \quad |X| = |Y| = 2k.$$

In coordinates $0, 1, 2$, the substrings are 010 and 202 . In coordinates $3 \leq j \leq 2k - 6$, both entries lie in alternating strings of opposite parity. In the last five coordinates the substrings are 10121 and 21202 . Thus $X_j \neq Y_j$ at every coordinate, so all diameters are proper.

The odd row covers every odd $n \geq 9$, the second row every $n \equiv 2 \pmod{4}$, and the last two rows cover 20 and every multiple of four from 24 onward. The only uncovered orders $n \geq 6$ are $7, 8, 12, 16$. $\square$

4. EXACT OBSTRUCTION AT THE FOUR EXCEPTIONAL ORDERS

Lemma 4.1 (cyclic run bound). *If a cyclic sign word has no three consecutive equal signs and contains p plus signs and q minus signs, then $p \leq 2q$ and $q \leq 2p$.*

Proof. If both signs occur, the runs of either sign have length at most two, and the runs of plus and minus alternate. Thus the total length of the plus runs is at most twice the number of minus runs, which is at most $2q$; the other inequality is symmetric. A one-sign word violates the stated run condition. $\square$

Proposition 4.2. *The following four graphs are not three-colourable:*

$$C_7^{(3)}, \quad C_8^{(3)}, \quad C_{12}^{(3)}, \quad C_{16}^{(3)}.$$

Proof. Suppose a colouring exists, and apply Lemma 2.2. If its difference word has p plus signs, closure gives

$$(5) \quad 2p - n \equiv 0 \pmod{3}.$$

For $n = 7$, equation (5) gives $p = 2$ or $p = 5$. In either case the majority sign occurs five times and the minority sign twice, contradicting Lemma 4.1.

For the three even orders, equation (5) first gives the possibilities in the middle column below. Lemma 4.1 leaves only the last column.

n	possibilities from (5)	after the run bound
8	1, 4, 7	4
12	0, 3, 6, 9, 12	6
16	2, 5, 8, 11, 14	8

Thus, writing $n = 2m$, exactly m of the $2m$ signs are plus.

Let q_i count plus signs in the cyclic length- m window beginning at i . The opposite window contains all remaining signs, so $q_{i+m} = m - q_i$. Here $m \in \{4, 6, 8\}$ is even, and $|q_{i+1} - q_i| \leq 1$. The integer sequence of window counts from i to $i + m$ must therefore pass through $m/2$. At such an index the window's sign sum is

$$q_i - (m - q_i) = 2q_i - m = 0,$$

contradicting the nonzero half-window condition in Lemma 2.2. Hence none of the four graphs is three-colourable. $\square$

Theorem 4.3. *For every integer $n \geq 6$,*

$$a(n) = 0 \iff n \in \{7, 8, 12, 16\}.$$

Proof. Proposition 3.1 supplies a proper three-colouring at every order outside the displayed set, while Proposition 4.2 excludes every order in it. $\square$

5. EXACT DIAGNOSTIC CERTIFICATE

The proof above is finite and hand-checkable; computation is not used to infer its infinite conclusion. The serialized data and verifier are stored under `certificates/` as

`builder_construction_spec.json`, `verify_builder_construction.py`.

The data serialize Definition 2.1, the two blocks, the four families, and the exception list. The verifier parses that input, reconstructs the colour word at each requested order, and checks every graph edge directly. Separately, it exhausts all 2^n adjacent difference words at $n = 7, 8, 12, 16$. From the project root, one may run

```
python3 certificates/verify_builder_construction.py
-max-n 10000
```

The archived run reports PASS, checks 9,991 constructed orders, and exhausts respectively 128, 256, 4096, 65536 sign words at the four exceptions. A second verifier in `verifier/verify_zero_set.py` uses a separately written construction and direct graph search. These checks are supplemental: Propositions 3.1 and 4.2 prove the theorem for all orders.

DATA, CODE, AND COMPUTATIONAL DISCLOSURE

The serialized certificates, deterministic Python verifiers, audit records, and release hashes accompany this manuscript. No random search is used by the released verification commands. OpenAI Codex assisted with literature search, proof discovery, exact checking, adversarial review, and typesetting; separate agent roles developed two different constructive proofs and an independent referee reconstructed the argument from the graph definition. All claims in the theorem are supported by the explicit human-readable proof above. No proof assistant (Lean, Coq, Isabelle, or otherwise) was used.

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